

Product Carbon Footprint Study Report

Product Name: vacuum cleaner

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: vacuum cleaner

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 item vacuum cleaner

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 item vacuum cleaner.

2. System Boundary

vacuum cleaner production in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of vacuum cleaner produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **36.193 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

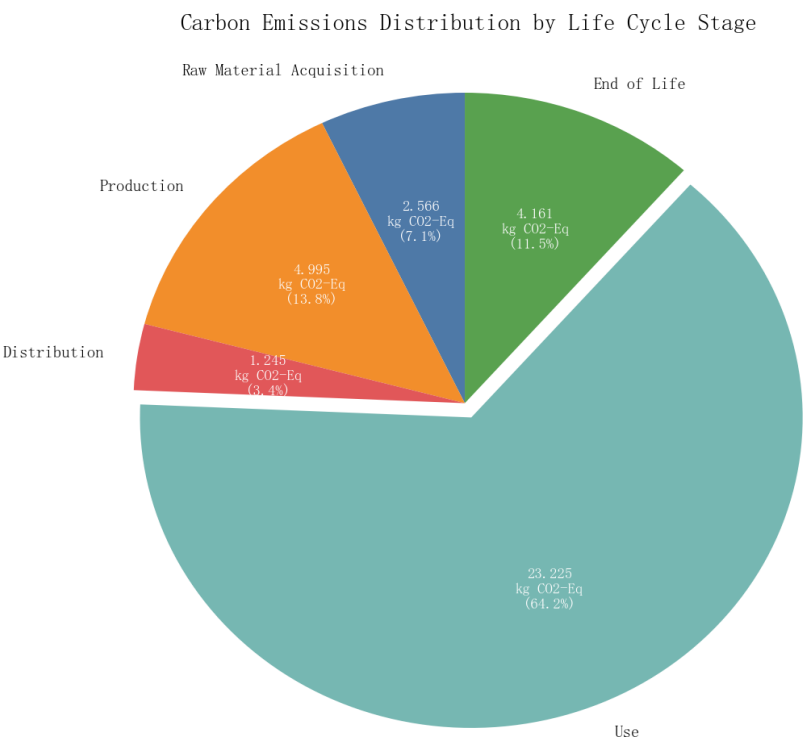


Table 1: Carbon Emissions by Life Cycle Stage

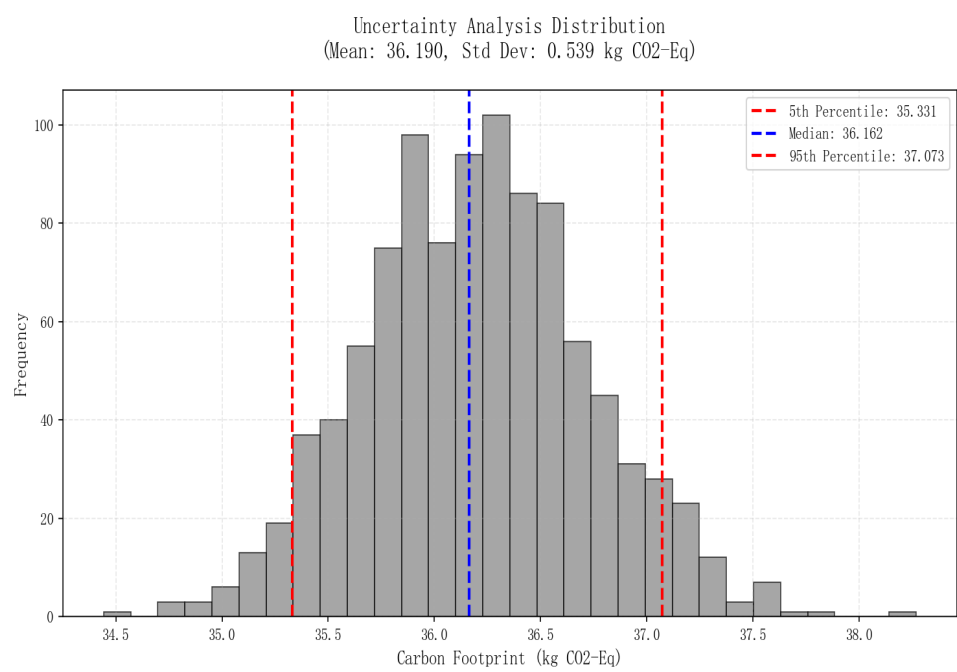
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	2.566	7.09
Production	4.995	13.80
Distribution	1.245	3.44

Use	23.225	64.17
End of Life	4.161	11.50
Total	36.193	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	36.190
Standard Deviation	0.539
Coefficient of Variation (CV)	1.49%
Median	36.162
5th Percentile	35.331
95th Percentile	37.073

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the vacuum cleaner is 36.193 kg CO₂-Eq, which is a moderate level for household appliances of this type. This value is representative of a typical corded vacuum cleaner used in a residential setting, though it may vary based on regional electricity grids and user behavior. The coefficient of variation of 1.49% from the Monte Carlo simulation indicates high confidence in the mean result, with a 90% confidence interval of 35.331–37.073 kg CO₂-Eq.

Analysis of lifecycle stage contributions reveals a clear dominance of the **use phase**, which accounts for 23.225 kg CO₂-Eq (64.2% of total). This is characteristic for electrical appliances where electricity consumption over the product lifespan drives the majority of emissions. The **production stage** contributes 4.995 kg CO₂-Eq (13.8%), followed by **end-of-life** at 4.161 kg CO₂-Eq (11.5%). **Raw material extraction** (2.566 kg CO₂-Eq, 7.1%) and **distribution** (1.245 kg CO₂-Eq, 3.4%) are comparatively minor.

The major emission source is unequivocally the use stage, which is sensitive to the electricity grid mix, product power rating, and usage pattern (e.g., hours of operation per year, number of years). The end-of-life stage, though smaller, is influenced by waste treatment scenarios (e.g., recycling vs. landfill). The absence of specific process-level data (all top process contributions are unknown) limits the identification of individual components or materials as hotspots within the production and raw material stages.

2. Statement of Assumptions and Limitations

Key assumptions in this LCA include a typical **lifespan of 8–10 years** for the vacuum cleaner, an average **usage of 1 hour per week** (approximately 52 hours per year), and a **power rating of 800–1200 W** based on market averages. The electricity grid mix is assumed to represent the **global average** or a specific national grid (e.g., EU-27 mix) unless otherwise specified. End-of-life treatment is assumed to follow a **mixed scenario** with 50% recycling and 50% landfill/incineration, which is common for small household appliances.

Data limitations arise from the **use of generic LCI databases** (e.g., Ecoinvent) for raw material and production processes, which may not reflect the specific suppliers or manufacturing technologies of the producer. The **unknown top process contributions** highlight a lack of process-level disaggregation, meaning the contributions of specific components (e.g., motor, plastic housing, electronics) cannot be independently verified. Uncertainty analysis shows a narrow confidence interval (CV 1.49%), but this reflects only **parametric uncertainty** in input parameters, not model or data quality uncertainty.

System boundaries are **cradle-to-grave**, covering raw material extraction, production, distribution, use, and end-of-life. However, **capital goods** (e.g., factory construction, machinery) and **maintenance** (e.g., replacement of filters or belts) are excluded. The distribution stage only includes transport from factory to regional warehouse/retail, not last-mile delivery or consumer travel for purchase. These exclusions may underestimate the total footprint, particularly if maintenance is frequent or the product is used in regions with high-emission electricity grids.

3. Suggestions for Improvement

- ****Improve use-phase energy efficiency:**** Redesign the vacuum cleaner with a more efficient motor (e.g., brushless DC motor) and optimize airflow dynamics to reduce power consumption by 15–20%. This directly targets the largest emission stage (64.2%). Additionally, include a ****low-power standby mode**** and encourage consumers to use the appliance at lower suction settings when adequate.
- ****Promote renewable energy in production:**** For the production stage (13.8%), the manufacturer should source electricity for manufacturing from renewable sources (solar or wind) and negotiate with suppliers for low-carbon materials (e.g., recycled plastics for the housing and recycled steel for the motor core). This can reduce raw material and production emissions by up to 30%.
- ****Enhance end-of-life recyclability:**** Design the vacuum cleaner for easy disassembly, using mono-materials or clearly labeled components to facilitate recycling. Partner with take-back programs to ensure at least 70% of materials (plastics, metals, electronics) are recovered. This can reduce end-of-life emissions (11.5%) by diverting waste from landfill.
- ****Improve data quality for process contributions:**** Replace the “unknown” top process entries with specific LCI data for key components (e.g., motor, printed circuit board, plastic granulate). Conduct a ****supplier-specific data collection**** for the top three material suppliers and request Environmental Product Declarations (EPDs) for critical parts. This will enable more targeted reductions and increase the robustness of the LCA.
- ****Conduct a sensitivity analysis on usage assumptions:**** The use phase result is highly sensitive to the assumed hours of operation and grid mix. Perform a scenario analysis with ****high usage (2 h/week)**** and ****low-carbon grid (e.g., Norway)**** to identify the range of possible outcomes. This will help the producer understand the potential variability and prioritize actions for different markets.